

2026 年 ACCA 全国就业力 X 未来商业创想大赛案例资料

Case study material for 2026 Job Hunting Competition THINK AHEAD

特此鸣谢案例合作伙伴

Special thanks to Case Study Partner

ESG绿碳智汇
GREEN CARBON INTELLIGENCE

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背景综述

在“双碳”战略与制造业高端化、智能化、绿色化转型的大背景下，企业正同时面临三重压力：

1）能源成本持续上升与用能结构复杂化；

2）监管与资本市场对 ESG 表现的要求日益严格；

3）数字化与智能化投入回报周期不确定。

绿碳智汇（宁波）科技发展有限公司（以下简称“绿碳智汇”）是一家以 AI 算法 + 能源系统工程 + ESG 咨询为核心能力的创新型企业，致力于为工业园区、高端制造企业和公共建筑提供一体化的能源优化与节能减碳解决方案。公司通过自研算法、数字孪生建模和智能控制平台，在更换部分硬件的前提下，帮助客户实现 10%–30% 的节能效果，并同步提升系统稳定性与合规水平。

然而，在实际商业化过程中，绿碳智汇发现：

- AI 能效优化和 ESG 价值往往需要中长期数据积累才能完全显现；
- 客户在决策时更关注 短期成本回收、现金流影响与财务绩效；
- 市场竞争者既包括传统能源管理公司，也包括咨询机构和平台型数字企业。

在此情境下，绿碳智汇需要重新审视自身的商业模式、竞争定位、组织与治理能力，并思考未来 5 年的可持续增长路径。

商业概览

在深入分析前，请通过下表快速了解绿碳智汇在案例发生时点（约 2025 年）的核心商业与财务特征（此概览是后续所有财务与战略分析的数据母体和逻辑起点，请在此基础上理解公司的机遇与挑战）：

表 1：绿碳智汇当前商业模式概览（案例发生时点）

维度	关键信息（示例）
商业模式	项目制为主，产品化探索中
年交付项目数	6–8 个
单项目周期	6–12 个月
客户回款方式	里程碑制（如 30%-40%-30%）
是否存在订阅收入	否 / 正在试点

如下附件 1-6 为案例补充材料，可结合如上企业背景和介绍一起参考。

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附件 1：行业背景介绍
附件 2：公司概况
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附件 6：市场竞争对手分析

假设你是绿碳智汇的战略咨询顾问，正在起草公司下一阶段发展的战略规划报告。该报告将为企业设计远景使命和价值观，并且设定未来中长期发展目标。在系统评估绿碳智汇面临的机遇，挑战，竞争优势及劣势之后，明确其业务布局与发展路径，设计可持续的运营能力及管理能力，从而实现经济价值与环境、社会价值的协同创造。

该报告要求如下：

任务一

在 AI 驱动的能效解决方案中，ESG 与能效价值往往需要依赖长期数据积累与模型优化才能逐步显现，而企业客户更关注短期成本回收与财务绩效。请分析这一现象产生的根本原因，并说明这一矛盾对绿碳智汇与客户决策过程意味着什么。

任务二

典型项目：绿碳智汇在 2026 年初中标了一个工业园区综合能源优化项目，合同总金额为 500 万元，预计项目周期为 8 个月。该项目是公司向产品化转型前的典型项目制业务。
已知成本与费用数据：

1. 直接成本（与项目直接相关）：
- 项目团队人力成本：合同金额的 40%
- 专用算法开发与数据采集成本：合同金额的 12%
- 差旅、现场实施等费用：合同金额的 8%

2. 间接费用分摊：
- 公司总部管理费用（行政、财务等）按项目收入的 10% 分摊
- 销售费用（获客成本）按项目收入的 8% 分摊

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要求：

1. 计算该项目的税后净利润。假设企业所得税率为 25%

2. 如果客户采用 40%-30%-30%的里程碑付款方式（签约 40%，中期 30%，验收 30%），且项目成本在 8 个月内均匀发生，请分析该项目对公司季度现金流的影响，并且提出现金流管理的主要风险点及缓释机制。

3. 然而，目前在市场中，合同能源管理的模式仍是主要的商业模式，请分析其相对于里程碑模式的优劣势，及企业如何管理相关的财务风险。

任务三

在当前绿色能效与 ESG 服务市场中，第三方解决方案提供商既包括传统能源管理企业、ESG 咨询机构，也包括数字化与平台型公司。因此，目前市场竞争非常激烈。另外，外部市场虽然有绿色转型的机遇存在，但也存在不少发展阻力和制约。请综合内外部情况，综合分析企业的差异化竞争力。

任务四

基于内外部竞争分析及企业愿景及目标，提出未来绿碳智汇的 5 年的业务发展模式，并提出规划方案，包括但不限于市场选择，竞争定位，客户选择，业务模式设计及业务组合等。在此基础上，请提供定量业务发展规划。

另外，并通过定性分析提出，为了实现这一业务发展目标，在运营及管理模式层面，企业应重点提升哪几方面能力，以及如何提升。

任务五

7 月 9 日北京商业创想训练营现场发布，敬请期待。

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附件 1：行业背景（公开渠道检索）

在“双碳”目标持续推进的背景下，企业正面临由监管、资本市场与社会责任要求共同驱动的可持续发展转型压力。低碳发展已不再局限于单一的能源管理问题，而逐步延伸至企业运营、治理、信息披露与价值链管理等多个层面。

一方面，制造业、产业园区及大型公共建筑在能源成本上升与用能结构复杂化的背景下面临运营压力；另一方面，监管机构、金融机构及核心客户对企业 ESG 表现的关注度不断提高，企业在环境绩效、治理能力及信息透明度方面的表现，正逐渐影响其融资条件、市场准入及长期竞争力。

在此过程中，传统以设备更新或单点工程改造为主的节能方式，逐渐暴露出投入高、周期长、难以持续优化等局限；而以报告编制或合规填报为主的部分 ESG 服务，也面临与企业实际运营脱节的问题。市场开始出现对能够同时连接技术落地、管理体系与合规要求的综合型解决方案的需求。

随着企业可持续发展相关需求的持续扩展，原本由企业内部承担的部分职能逐步向外部专业机构转移，形成了多元化的第三方服务商生态。从业务类型看，当前市场中的第三方服务商大致包括：以工程实施和节能改造为主的技术服务商；以碳盘查、ESG 报告及合规支持为主的咨询与审计机构；以及以数字化与算法能力为核心，强调通过系统建模与智能控制提升运行效率的综合服务商。

不同类型服务商在服务边界、技术投入、商业模式及风险承担方式等方面存在显著差异，企业在选择合作方时，通常需要在成本控制、实施周期、效果可持续性以及合规要求之间进行综合权衡。

行业新闻
2025 年全国碳市场政策加速推进。中共中央办公厅和国务院办公厅于 2025 年 8 月联合印发《关于推进绿色低碳转型加强全国碳市场建设的意见》，明确扩大碳市场覆盖行业，新增排放量约 30 亿吨，占全国 CO_2排放量 20% 以上。企业未来将面临更严格的排放成本与市场约束。 *——腾讯新闻*
2025 年全国节能宣传周报道指出，全国多行业正推动极致能效改造，钢铁等高耗能行业深入实施节能装备升级与效率提升行动，为能源密集型产业的绿色转型提供现实路径。 *——中国钢铁新闻网*

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“鼓励公共机构推行能源托管服务。” *——国务院关于加快建立健全绿色低碳循环发展经济体系的指导意见*
根据市场研究机构报道，全球 **AI 能效工具市场预计将以约 34.1% 的年复合增长率增长，到 2034 年规模达约 1636 亿美元**，表明 AI 在能源系统优化领域具有高速增长和广泛采用趋势。 *——公开报道*

附件 2：公司概况（绿碳智汇的发展阶段与业务定位）

绿碳智汇（宁波）科技发展有限公司成立于 2025 年年初，紧密响应国家“双碳”战略及能源政策的发展趋势，是一家聚焦 AI 算法驱动的能源系统优化与 节能减碳绩效管理 的技术型初创企业。公司核心团队来自能源工程、自动化控制、数据算法与可持续发展等多个领域，致力于通过数字化和智能化手段，提升复杂能源系统的运行效率。

截至案例发生时点，公司已在若干工业企业、园区及公共建筑场景中完成项目落地，技术路径在实际运行中得到验证，但公司整体仍处于 “技术成熟度提升、商业模式探索中” 的关键发展阶段。

截止 2025 年年底，案例企业的年度总营收 5000 万，收入结构参考材料三。
另外，根据跟创始人的访谈，企业的愿景是“智驭能源，惠及万企”。就财务目标来看，企业希望未来 3 年，企业实现收入的翻番，利润上一大台阶。

附件 3：主营业务与服务内容

3.1 服务模式概述

绿碳智汇公司的服务主要围绕“双碳领域，ESG 领域，能源业务”三个方向展开。

1.双碳领域：

- 碳盘查与 LCA 计算

通过碳盘查与生命周期评估（LCA）服务，帮助企业识别并量化碳排放，提供减排方案，助力绿色低碳转型。

- 定制化低碳咨询

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提供量身定制的低碳咨询服务，涵盖供应链优化、技术改造等领域，推动绿色发展。

- 碳配额交易市场咨询

提供碳交易市场的政策分析与操作指导，帮助企业合理管理碳资产，提升市场竞争力。

- CDP 与 CBAM 填报

为企业提供碳披露项目（CDP）和欧盟碳边境调整机制（CBAM）的填报指导，提升透明度和合规性。

- CCER 开发

开发并管理符合国家标准的 CCER 项目，为企业提供自愿减排交易渠道，促进低碳发展。

- 绿色金融服务

提供绿色金融资源和融资支持，助力企业绿色转型，实现长效共赢。

2.ESG 领域：

- ESG 咨询指导

为企业提供全面的 ESG 咨询，帮助优化 ESG 表现，提升社会责任认可度和投资吸引力。

- ESG 管理体系建立

协助企业建立和完善 ESG 管理体系，将 ESG 战略转化为实际行动，提升治理水平。

- ESG 报告撰写

编写符合国际标准的 ESG 报告，展示企业在环境、社会和治理领域的投入与成效。

- ESG 管理拓展业务

拓展 ESG 管理业务，提供创新解决方案，支持企业在多元化领域深化 ESG 战略。

3.能源业务：

- 搭建产学研合作平台

建立合作平台，推动节能技术落地与科研成果转化，助力企业提高能源利用率、降低成本，支持低碳发展。

- 开展行业研讨分享会

定期举办研讨会，分享节能新技术、行业趋势与政策，推动企业创新管理与技术应用，降低运营成本。

- 推动行业标准实施

推动节能标准制定，提供标准化建议。

- 新能源与绿色能源的应用

推广绿色能源解决方案，帮助企业在节能降耗方面实现转型，提升能源效率，减少能源成本，推动绿色低碳发展。

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在上述业务框架下，以下内容将以能源业务为例，对相关技术模式、应用方式及实践情况进行进一步说明。

绿碳智汇当前提供的能源服务主要聚焦于以下几个核心领域：

1. 能源系统优化：通过引入先进的人工智能技术，实现能源系统的智能调度、运 行优化及管理升级，从而显著提升能源利用率与系统效率。

2. 节能诊断与技术改造：针对企业现有的能源设施与生产流程，开展深入评估与 节能诊断，依据实际情况制定针对性的技术改造与节能优化方案，以降低能源 消耗和运营成本。

3. 用能数据分析与管理：依托企业能源消耗数据，运用大数据分析和机器学习模型，精准构建能效分析与预测模型，为企业提供科学的决策支持，推动能源管理的智能化。

4. 低碳能源转型咨询：为企业量身定制低碳能源转型战略，助力企业实现绿色可持续发展，推动可再生能源、清洁能源技术的应用与产业化，促进企业向低碳经济转型。

相关服务通常以项目形式展开，项目周期根据系统复杂程度与数据积累情况存在差异。

其中，能源方面主要技术优势包括：

1. 优化设计与建模：通过全面建模和优化算法，提高能源配置效率，降低能耗。
2. 智能优化控制：实时动态调整设备运行，提升能源利用效率。
3. AI 与机器学习应用：跨学科技术融合，提供定制化节能解决方案。
4. 节能减排成效：项目普遍达成了 30%+ 的碳减排和显著的电费节省。
5. 无缝集成与市场应用：技术解决方案可与现有系统无缝对接，助力企业实现降本增效。

这些技术已广泛应用于工业、建筑及企业园区，助力可持续发展与绿色低碳目标。具体而言的技术模式包括：

1.低碳设计与优化

通过构建能源系统模型，对建筑与园区的用能结构进行低碳化设计与运行优化，在满足功能需求的前提下，系统性降低整体碳排放与能源浪费。于宁波市政务服务中心园区落地，对其综合能源系统进行低碳设计与优化，完成低碳园区与关键技术的应用示范。实现 30%碳减排和提升 50%用电柔性。

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2.AI 与优化算法

基于高精度数据模型与 AI 算法，对空调等关键用能系统进行持续优化调度，实现能效提升，并将节能效果转化为可量化、可验证的数据成果。于舍弗勒（中国）落地，舍弗勒集团作为工业产品和汽车国际供应商，其中国工厂能源消耗巨大，尤其是制冷空调系统。项目组建跨学科团队，运用多种 AI 算法优化能耗模式，实现经济利益和节能减排的双赢。实现节电约 5000 度/日，每年节约约 120 万元。

3.工业能效优化

结合机理仿真与机器学习方法，对工业暖通与复杂用能系统进行协同优化，在保障生产稳定性的同时，提高系统运行效率并降低长期能耗成本。于上海叠腾科技落地，将人工智能应用于工业能源数字化领域，助力其获得数千万元 Pre-A 轮融资同时实现提升 10%能效，每年节省数百万能源费用。针对这些企业工业暖通系统的高能耗问题，提供定制化解决方案，实现能耗精准管理。产品可无缝集成进企业现有的暖通和能源管理系统，满足不同企业的降本节能需求，目前已在国内光伏、锂电、面板行业龙头企业实现应用。

4.动态优化控制

通过动态控制策略实时调节除湿与能源系统运行状态，使系统能够响应负载与环境变化，在不增加硬件投入的情况下实现能效与成本的双重改善。于某锂电企业落地，项目团队与多家锂电企业合作，针对锂电池生产对环境湿度的严格要求，开发了一款优化除湿系统。经过定制化安装和调试，与企业现有系统无缝对接。实现提升约 10%能源效率，每年节省百万电费。

5.智能优化与强化学习

利用强化学习等智能优化技术，让系统在持续运行中不断“学习”最优控制策略，从而逐步减少空调等设备的能耗并提升整体运行表现。于卡耐基梅隆大学落地。美国卡耐基梅隆大学的智能办公空间是一个具有创新性和前瞻性的项目，聚焦于智能办公场景，面积约 7000 平方英尺。原有门子控制逻辑已不适应现代能源高效利用需求，存在优化空间。项目组运用前沿智能技术与数据分析手段，大幅减少能耗，降低运营成本，打造绿色高效智能办公空间。实现节省 16%空调能耗。

6.建筑能源优化

基于预测控制模型，对建筑能源系统进行前瞻性调度与优化，在降低空调能耗的同时减少设备损耗，提升系统稳定性与使用寿命。于美国 Phipps 温室落地，作为菲普斯温室植物园的绿色建筑典范，致力于通过全面调研和数据采集优化能源利用，集成能源模型与楼宇自动化系统，

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实现节能降耗，同时满足多样化需求，为可持续景观建筑能源利用树立典范。节省 20%空调能耗，延长设备寿命。

另外，能源合同管理是目前行业内主要的商业模式，具体的商业模式如下：

项目要素	具体说明
合作模式	合同能源管理（EMC）：服务商投资并建设
甲方投入与职责	牵头协调内部资源，配合完成前期能耗数据采集与接入
甲方节能收益	直接分享项目年节能收益的30%，预计每年获得收益约 84万元（基于年节电收益280万元据实结算）
合作期限	6年（合作期满后，平台资产及所有权可按约定移交甲方）
提前买断权	合作期内，甲方享有灵活的提前买断选择权：提前买断价格 = 剩余合作年限 × 当年实际节能收益 × 70% × 折现系数（0.6-0.8）
核心价值与收益	直接经济收益、管理升级、ESG战略价值、政府项目申报

3.2 收入结构

公司当前收入主要来源于以下几类：

表 2：绿碳智汇当前收入结构（截止 2025 年年底）

收入类型	占比	收入特征
定制化能效优化项目	65%	一次性收入，毛利高但不可复制
ESG / 双碳咨询	20%	项目制，交付依赖人力
软件/平台试点服务	10%	订阅制，尚未规模化
其他（培训、研讨等）	5%	非核心

附件 4：客户画像与应用场景

4.1 客户类型分布

根据已落地项目情况，绿碳智汇的客户主要集中在以下几类场景：

表 3：客户类型分布

客户类型	客户数量占比	收入贡献占比
高端制造企业	40%	55%
工业园区	30%	30%
公共建筑/机构	30%	15%

附件 5：国际市场与出海相关背景

在国内市场之外，随着全球范围内碳减排政策与 ESG 披露要求的不断强化，能源效率提升与低碳技术服务在国际市场上的需求亦逐步显现。然而，相较于国内环境，海外市场在监管体系、数据合规、技术标准及商业惯例等方面存在明显差异。

对第三方能源与低碳服务提供商而言，出海不仅意味着市场机会的拓展，也伴随着数据跨境、模型适配、本地法规合规及责任界定等方面的挑战。因此，企业在考虑国际化布局时，通常需要在技术可复制性、组织能力匹配度以及风险可控性之间进行综合权衡。

附件 6：市场竞争对手

1. 工业自动化与数字化巨头

主要优势：品牌、全栈产品线、庞大的客户基础与渠道。他们能提供从传感器、PLC、变频器到 MES、云的完整数字化方案。

商业模式：销售硬件产品与大型软件平台许可，通过系统集成商交付。

典型客户：大型国企、高端制造、新建或大型改造项目。

2. 电网公司与综合能源服务商

主要优势：极强的客户信任、能源数据入口、资金与金融手段。擅长运营和投资。

商业模式：能源费用托管（合同能源管理）是主流，做投资和运营，有时也集成外部产品。

典型客户：园区、公共建筑、高耗能企业。

3. AI 软件与 SaaS 平台公司

主要优势：优秀的算法模型、互联网产品体验、灵活的软件开发能力。

商业模式：提供云平台 SaaS 服务，按点位或服务订阅收费。

典型客户：有一定信息化基础、注重数据的中型企业。

4. 传统的工程集成商与节能服务公司

主要优势：本地化服务、客户关系、灵活的项目定制能力。

商业模式：项目制，采购硬件、设计编程、施工调试，赚取工程与产品差价。

典型客户：各类有节能改造预算的企业。

5. 其他公司（自行检索）

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Case Description

Under the background of the “dual-carbon” strategy and the high-end, intelligent, and green transformation of the manufacturing industry, enterprises are simultaneously facing triple pressures:

1) The persistent rise in energy costs and the increasing complexity of the energy consumption structure.
2) Regulatory and capital market requirements for ESG performance are becoming increasingly stringent.
3) The return on investment for digitalization and intelligentization initiatives is subject to uncertainty.

ESG Green Carbon Intelligence (Ningbo) Co., Ltd (hereinafter referred to as “ESG GCI”) is an innovative enterprise with core competencies in AI algorithms + energy systems engineering + ESG consulting, and is committed to providing integrated energy optimization and energy-saving carbon reduction solutions for industrial parks, high-end manufacturing enterprises, and public buildings. Through proprietary algorithms, digital twin modelling, and intelligent control platforms, the company enables clients to achieve energy savings of 10%-30 % with minimal hardware replacement, while simultaneously enhancing system stability and compliance standards.

However, during the actual commercialization process, ESG GCI observed that:

- AI energy efficiency optimization and ESG value often require medium- to long-term data accumulation to be fully manifested;
- Clients prioritize short-term cost recovery, cash flow impact and financial performance when making decisions;
- Market competitors encompass traditional energy management firms, consultancy organizations and platform-based digital enterprises.

In this context, ESG GCI must reassess its business model, competitive positioning, organizational and governance capabilities, and contemplate a sustainable growth trajectory for the next five years.

Business Snapshot

Before delving into the in-depth analysis, please quickly understand the core business and financial characteristics of ESG GCI at the case occurrence point (approximately 2025) through the following table (this overview serves as the data parent and logical starting point for all subsequent financial and strategic analyses. Please understand the company's opportunities and challenges based on this):

Table 1: Overview of Current Business Model of ESG GCI (Case Occurrence Time Point)

Dimension	Key Information (Example)
Business model	Project-based approach as the primary focus, with ongoing exploration into productisation.
Annual Project Deliveries	6-8
Single project cycle	6-12 months
Customer Payment Method	Milestone-based (e.g., 30%-40%-30%)
Does Subscription Revenue Exist	No / Pilot in progress

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The supplementary materials for the case study are provided in Appendices 1-6 below. You can review them along with the company background and introduction provided earlier.

Appendix 1: Industry Background Overview

Appendix 2: Company Profile

Appendix 3: Core Business Operations and Service Offerings

Appendix 4: Customer Profiles and Application Scenarios

Appendix 5: International Market and Overseas Expansion Context

Appendix 6: Competitor Analysis

Assume you are a strategic consultant for ESG GCI, drafting the strategic planning report for the company's next phase of development. This report will define the organization's vision, mission, and values, while establishing medium- to long-term development objectives. Following a systematic assessment of ESG GCI's opportunities, challenges, competitive strengths and weaknesses, it will clarify its business layout and development pathways, designing sustainable operational and management capabilities to achieve the synergistic creation of economic value alongside environmental and social value.
The report requirements are as follows:

Task I

Within AI-driven energy efficiency solutions, the value of ESG and energy efficiency often requires long-term data accumulation and model optimisation to gradually manifest, whereas corporate clients prioritise short-term cost recovery and financial performance. Please analyse the fundamental causes of this phenomenon and explain what this contradiction implies for ESG GCI and the client decision-making process.

Task II

Typical Project: In early 2026, ESG GCI secured a contract for an industrial park comprehensive energy optimisation project, with a total contract value of RMB 5 million and an anticipated project duration of eight months. This project represents a typical project-based operation undertaken by the company prior to its transition towards productisation.
Known Cost and Expense Data:

1. Direct costs (related to the project directly):
- Project team personnel costs: 40% of the contract amount
- Special algorithm development and data collection costs: 12% of the contract amount
- Travel, on-site implementation, etc.: 8% of the contract amount

2. Indirect costs apportion:
- Company headquarters management expenses (administration, finance, etc.) are apportioned at 10% of the project revenue
- Sales expenses (acquisition costs) are apportioned at 8% of the project revenue

Requirements:
1. Calculate the project's net profit after tax. Assume a corporate income tax rate of 25%.
2. Assuming the client adopts a 40%-30%-30% milestone payment structure (40% upon

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contract signing, 30% at mid-term, 30% upon acceptance), and project costs are incurred evenly over 8 months, analyse the project's impact on the company's quarterly cash flow. Identify key cash flow management risks and propose mitigation mechanisms.

3. However, the Energy Performance Contracting model remains the predominant business model in the current market. Please analyse its advantages and disadvantages relative to the Milestone Model, and how enterprises should manage the associated financial risks.

Task III

Within the current green energy efficiency and ESG services market, third-party solution providers encompass traditional energy management enterprises, ESG consultancies, as well as digital and platform-based companies. Consequently, market competition is exceptionally fierce at present. Furthermore, while external markets present opportunities for green transformation, they also harbour numerous developmental obstacles and constraints. Please conduct a comprehensive analysis of the company's differentiated competitive strengths, taking into account both internal and external factors.

Task IV

Based on an analysis of internal and external competitive factors alongside the company's vision and objectives, propose a five-year business development model for Future ESG GCI. This should include a planning framework encompassing, but not limited to, market selection, competitive positioning, customer targeting, business model design, and portfolio strategy. Building upon this foundation, provide a quantitative business development plan. Furthermore, through qualitative analysis, identify the key operational and managerial capabilities the company must enhance to achieve these business development objectives, along with specific approaches for their improvement.

Task V

To be unveiled at the THINK AHEAD training camp in Beijing on 9 July. Stay tuned.

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Appendix 1: Industry Background Overview (search via public channels)

Against the backdrop of advancing “dual carbon” goal, enterprises face mounting pressure to transition towards sustainable development, driven by regulatory requirements, capital market and social responsibility imperatives. Low-carbon development is no longer confined to energy management alone, but is progressively extending across multiple dimensions, including business operations, governance, information disclosure and value chain management.

On the one hand, manufacturing industries, industrial parks and large public buildings face operational pressures amid rising energy costs and increasingly complex energy consumption structures; on the other hand, regulators, financial institutions and core clients are paying increasing attention to the ESG performance of enterprises, and the performance of enterprises in terms of environmental performance, governance capacity and information transparency is gradually affecting their financing terms, market access and long-term competitiveness.

In this process, traditional energy-saving approaches centred on equipment upgrades or single-point engineering modification have progressively revealed limitations such as high investment costs, lengthy implementation cycles, and difficulties in achieving sustained optimization. Meanwhile, certain ESG services primarily focused on report compilation or compliance reporting face the challenge of being disconnected from actual business operations. The market is now witnessing a demand for integrated solutions capable of simultaneously bridging technological implementation, management systems, and regulatory requirements.

As the scope of corporate sustainability requirements continues to expand, certain functions traditionally handled internally are progressively being outsourced to external specialist organizations, fostering a diverse ecosystem of third-party service providers. By business type, third-party service providers in the current market broadly encompass: technical service providers specializing in engineering implementation and energy-saving retrofits; consultancy and auditing firms focusing on carbon inventory, ESG reporting and compliance support; and integrated service providers centred on digital and algorithmic capabilities, emphasizing enhanced operational efficiency through system modelling and intelligent control.

Different types of service providers exhibit significant variations in service boundaries, technological investment, business models, and risk-bearing approaches. When selecting partners, enterprises typically need to weigh up a range of factors, including cost control, implementation timelines, sustainability of outcomes, and compliance requirements.

Industry News
National carbon market policies accelerated in 2025. In August 2025, the General Offices of the CPC Central Committee and the General Office of the State Council jointly issued the Opinions on Advancing Green and Low-Carbon Transformation and Strengthening the National Carbon Market, clarifying plans to expand the carbon market's coverage to additional sectors. This expansion will incorporate approximately 3 billion tonnes of new emissions, accounting for over 20% of the

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nation's total CO_2emissions. Enterprises will henceforth face stricter emission costs and market constraints. ——*Tencent News*
The 2025 National Energy Conservation Awareness Week report highlighted that multiple sectors nationwide are advancing extreme energy efficiency upgrades. High-energy-consuming industries such as steel are implementing comprehensive energy-saving equipment modernization and efficiency enhancement initiatives, thereby providing tangible pathways for the green transformation of energy-intensive industries. ——*China Steel News*
"Encourage public organizations to implement energy hosting services." ——*The State Council's Guidance on Accelerating the Establishment of a Sound Green, Low-Carbon and Circular Economic System*
According to market research firms, the global **AI energy efficiency tools market is expected to grow at a CAGR of approximately 34.1%, reaching approximately $163.6 billion by 2034**, indicating a trend of high growth and widespread adoption of AI in the field of energy system optimization. ——public reporting

Appendix 2: Company Profile (ESG GCI's Development Stages and Business Positioning)

ESG Green Carbon Intelligence (Ningbo) Co., Ltd was established in early 2025. Closely aligning with the national "dual carbon" strategy and evolving energy policy trends, it is a technology-driven start-up specialising in AI algorithm-driven energy system optimisation and energy-saving/carbon reduction performance management. The company's core team comprises experts from diverse fields including energy engineering, automation control, data algorithms, and sustainable development. They are dedicated to enhancing the operational efficiency of complex energy systems through digitalisation and intelligent solutions.

By the time of this case study, the company had successfully implemented projects across multiple industrial enterprises, industrial parks, and public building scenarios, with its technical approach validated in practical operation. However, the company remains in a critical developmental phase characterised by "enhancing technological maturity and exploring business models".

By the end of 2025, the case company's annual total revenue reached 50 million yuan, with the income structure detailed in Appendix 3.
Furthermore, according to interviews with the founder, the company's vision is "Intelligent Energy Management, Benefiting Enterprises Nationwide". Regarding financial objectives, the company aims to double its revenue and achieve a significant increase in profits within the next three years.

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Appendix 3: Principal Business Activities and Service Offerings

3.1 Overview of Service Models
ESG GCI's services primarily focus on three key areas: the dual carbon sector, the ESG sector, and energy operations.

1. Dual Carbon Sector:

- Carbon inventory and LCA calculations

Through carbon inventory and life cycle assessment (LCA) services, we assist enterprises in identifying and quantifying carbon emissions, providing reduction solutions to facilitate green and low-carbon transformation.

- Customized low-carbon advisory

Provide customized low-carbon advisory services covering areas such as supply chain optimization and technological upgrades to drive green development.

- Carbon allowance trading market advisory

Provide policy analysis and operational guidance for carbon trading markets, assisting enterprises in the prudent management of carbon assets to enhance market competitiveness.

- CDP and CBAM filing

Provide guidance to enterprises on completing the Carbon Disclosure Project (CDP) and the EU Carbon Border Adjustment Mechanism (CBAM) reporting requirements, enhancing transparency and compliance.

- CCER development

Develop and manage CCER projects that meet national standards, provide enterprises with voluntary emission reduction trading channels, and promote low-carbon development.

- Green financial services

Provide green financial resources and financing support to help enterprises in green transformation and realize long-term win-win situation.

2. ESG Sector:

- ESG advisory guidance

Provide comprehensive ESG consulting to help companies optimize their ESG performance and enhance their social responsibility recognition and investment attractiveness.

- ESG management system established

Assist companies in establishing and improving their ESG management systems, translating ESG strategies into practical actions, and enhancing their governance levels.

- ESG report writing

Produce an ESG report that meets international standards and demonstrates the company's commitment to and effectiveness in the area of environmental, social and governance.

- ESG management expansion business

We are expanding our ESG management business and providing innovative solutions to support companies in deepening their ESG strategies in diversified areas.

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3. Energy Operations:

- Building a platform for industry-university-research collaboration

Establishing a cooperation platform to promote the implementation of energy-saving technologies and the transformation of scientific research results, helping enterprises to improve energy utilization, reduce costs and support low-carbon development.

- Conducting industry seminar sharing sessions

Regular seminars are held to share new energy-saving technologies, industry trends and policies, and to promote innovative management and technology application in enterprises to reduce operating costs.

- Promoting the implementation of industry standards

Promote the development of energy-saving standards and provide standardization advice.

- New and green energy applications

Promote green energy solutions, help enterprises to realize transformation in energy saving and consumption reduction, enhance energy efficiency, reduce energy costs, and promote green and low-carbon development.

Within the aforementioned business framework, the following content will use the energy sector as an example to provide further clarification on the relevant technical models, application methods, and practical implementation.

ESG GCI currently provides energy services primarily focused on the following core areas:

1.Energy system optimization: by incorporating advanced artificial intelligence technologies, intelligent scheduling, operational optimization and management upgrades of energy systems are achieved, thereby significantly enhancing energy utilization rates and system efficiency.
2.Energy efficiency diagnosis and technical upgrades: conducting thorough assessments and energy efficiency diagnostics of existing energy infrastructure and production processes within enterprises. Developing tailored technical upgrade and energy optimization plans based on actual conditions to reduce energy consumption and operational costs.
3.Energy data analysis and management: leveraging corporate energy consumption data, employing big data analysis and machine learning models to construct precise energy efficiency analysis and forecasting frameworks. This provides enterprises with scientifically grounded decision support, driving the intelligent advancement of energy management.
4.Low-carbon energy transition advisory: tailoring bespoke low-carbon energy transition strategies for enterprises to facilitate green and sustainable development. This drives the application and industrialization of renewable and clean energy technologies, thereby advancing corporate transformation towards a low-carbon economy.

Related services are typically delivered on a project basis, with project timelines varying according to system complexity and data accumulation requirements.

Key technological advantages in the energy sector include:
1. Optimised design and modelling: Enhancing energy allocation efficiency and reducing consumption through comprehensive modelling and optimisation algorithms.

2. Intelligent optimisation control: Dynamically adjusting equipment operation in real time to improve energy utilisation efficiency.

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3. AI and machine learning applications: Interdisciplinary technology integration delivers bespoke energy-saving solutions.

4. Energy conservation and emission reduction outcomes: Projects consistently achieve over 30% carbon reduction alongside substantial electricity cost savings.

5. Seamless integration and market deployment: Technical solutions integrate seamlessly with existing systems, empowering enterprises to reduce costs and enhance efficiency.

These technologies have been extensively deployed across industrial, construction and corporate campus sectors, supporting sustainable development and green, low-carbon objectives. Specific technical approaches include:

1. Low-carbon design and optimization

By constructing energy system models, low-carbon design and operational optimization of energy consumption structures in buildings and campuses are implemented. This systematically reduces overall carbon emissions and energy wastage while meeting functional requirements. Implemented at the Ningbo Municipal Government Service Centre campus, the project involves low-carbon design and optimization of its integrated energy system, achieving demonstration applications of low-carbon campuses and key technologies. This delivers 30% carbon reduction and enhances electricity flexibility by 50%.

2. AI and optimization algorithms

Based on high-precision data models and AI algorithms, continuous optimization scheduling is implemented for key energy-consuming systems, such as air conditioning, to achieve enhanced energy efficiency. The resulting energy savings are converted into quantifiable, verifiable data outcomes. Implemented at Schaeffler (China), the Schaeffler—a global supplier of industrial products and automotive components—operates factories in China with substantial energy consumption, particularly within refrigeration and air conditioning systems. The project assembled a multidisciplinary team that employed multiple AI algorithms to optimize energy consumption patterns, achieving a dual win of both economic benefits and energy conservation. This resulted in daily electricity savings of approximately 5,000 kilowatt-hours, translating to annual savings of around 1.2 million yuan.

3. Industrial energy efficiency optimization

By integrating mechanism simulation with machine learning methodologies, we achieve synergistic optimization of industrial HVAC and complex energy-consuming systems. This approach enhances operational efficiency and reduces long-term energy expenditure while safeguarding production stability. Implemented by Shanghai Data Turing, this application of artificial intelligence within industrial energy digitalization secured tens of millions of yuan in Pre-A round financing. Concurrently, it delivered a 10% improvement in energy efficiency, yielding annual energy cost savings of millions. Addressing the high energy consumption of industrial HVAC systems in these enterprises, we deliver bespoke solutions enabling precise energy management. Our products integrate seamlessly into existing HVAC and energy

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management systems, meeting diverse cost-reduction and energy-saving requirements. They are currently deployed across leading enterprises in China's photovoltaic, lithium battery, and panel manufacturing sectors.

4. Dynamic optimized control

By dynamically adjusting the dehumidification and energy system's operational status in real time, the system responds to load and environmental changes, achieving dual improvements in energy efficiency and cost savings without additional hardware investment. Implemented at a lithium battery enterprise, the project team collaborated with multiple lithium battery manufacturers to develop an optimized dehumidification system tailored to the stringent environmental humidity requirements of lithium battery production. Following customized installation and commissioning, it integrated seamlessly with the enterprise's existing systems. This achieved an approximate 10% increase in energy efficiency, yielding annual electricity cost savings of one million yuan.

5. Intelligent optimization and reinforcement learning

Employing intelligent optimization techniques such as reinforcement learning, the system continuously "learns" optimal control strategies during ongoing operation. This progressively reduces energy consumption in equipment like air conditioning units while enhancing overall operational performance. Implemented at Carnegie Mellon University. Carnegie Mellon University's intelligent office space is an innovative and forward-thinking project focusing on smart office environments, spanning approximately 7,000 square feet. The existing Siemens control logic no longer meets modern energy efficiency requirements, presenting significant optimization potential. The project team employed cutting-edge intelligent technologies and data analysis to substantially reduce energy consumption and operational costs, creating a green, efficient, and intelligent workspace. Achieved a 16% reduction in air conditioning energy consumption.

6. Building energy optimization

Based on predictive control modelling, the building energy system undergoes forward-looking scheduling and optimization. This reduces air conditioning energy consumption while minimizing equipment wear, enhancing system stability and extending service life. Implemented at the Phipps in the United States, this project stands as a green building exemplar for the Phipps Conservatory and Botanical Gardens. It is dedicated to optimizing energy utilization through comprehensive research and data collection, integrating energy modelling with building automation systems to achieve energy savings and reduced consumption while meeting diverse operational demands. This establishes a benchmark for sustainable energy utilization in landscape architecture, achieving a 20% reduction in air conditioning energy consumption and extending equipment lifespan.

Additionally, energy contract management represents the primary business model within the industry at present. The specific business model is as follows:

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Item	Description
Cooperation Model	Energy Management Contract (EMC): Service provider handles investment and construction
Contributions and Responsibilities of Party A	Take the lead in coordinating internal resources to facilitate the completion of preliminary energy consumption data collection and integration
Party A's Energy-Saving Benefits	Directly share 30% of the project's annual energy-saving revenue, with projected annual earnings of approximately ¥ 840,000 (based on actual settlement of ¥ 2.8 million in annual electricity savings)
Term of cooperation	6 years (upon expiry of the cooperation period, the platform assets and ownership may be transferred to Party A as agreed)
Advance purchase rights	During the cooperation period, Party A shall enjoy a flexible early buyout option: Early buyout price = Remaining cooperation period × Actual energy savings for the year × 70% × Discount factor (0.6–0.8)
Core Values and Benefits	Direct economic benefits, management enhancement, ESG strategic value, government project applications

3.2 Income structure

The Company's current revenues are derived from the following categories:

Table 2: Current revenue structure of ESG GCI (at the time of the case occurrence)

Revenue type	Percentage	Revenue characteristics
Customized energy efficiency optimization projects	65%	One-time income, high gross profit but not replicable
ESG / Carbon Neutrality Consulting	20%	Project-based, delivery dependent on human resources
Software/Platform Pilot Services	10%	Subscription-based, not yet scaled
Others (training, seminars, etc.)	5%	Non-core

Appendix 4: Client Profiles and Application Scenarios

4.1 Distribution of client types

According to the projects that have already been implemented, ESG GCI's clients mainly focus on the following types of scenarios:

Table 3: Distribution of Client Types

Client type	% of total clients	% of total revenue
High-end manufacturing enterprises	40%	55%
Industrial parks	30%	30%
Public buildings/institutions	30%	15%

Appendix 5: International Markets and Overseas Related Background

Outside of the domestic market, as carbon emission reduction policies and ESG disclosure requirements continue to be strengthened globally, the demand for energy efficiency improvement and low-carbon technology services is also gradually emerging in the international market. However, compared with the domestic environment, there are obvious differences in the regulatory system, data compliance, technical standards and business practices in overseas markets.

For third-party energy and low-carbon service providers, going overseas not only means expanding market opportunities, but also poses challenges in terms of cross-border data, model adaptation, local regulatory compliance and liability definition. Therefore, when considering the international layout, companies usually need to make a comprehensive balance between the replicability of technology, the matching of organizational capabilities, and the controllability of risks.

Appendix 6: Market Competitors

1. Industrial Automation and Digitalisation Giants

Key Advantages: Brand recognition, comprehensive product portfolios, extensive customer bases and distribution channels. Capable of delivering end-to-end digital solutions spanning sensors, PLCs, variable frequency drives to MES and cloud platforms.
Business Model: Hardware sales and large-scale software platform licensing, delivered through system integrators.
Typical Clients: Large state-owned enterprises, high-end manufacturing, new construction or major retrofit projects.

2. Grid Operators and Integrated Energy Service Providers

Key strengths: Exceptional client trust, access to energy data, capital and financial instruments. Proficient in operations and investment.
Business model: Energy cost management (Energy Performance Contracting) is predominant, involving investment and operations, occasionally integrating external products.
Typical clients: Industrial parks, public buildings, high-energy-consumption enterprises.

3. AI Software and SaaS Platform Companies

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Key strengths: Superior algorithmic models, internet-grade product experience, agile software development capabilities.
Business model: Cloud-based SaaS services charged per point or subscription.
Typical clients: Medium-sized enterprises with established IT infrastructure and data-centric focus.

4. Traditional Engineering Integrators and Energy Service Companies

Key strengths: Localised service delivery, established client relationships, flexible project customisation capabilities.
Business model: Project-based approach involving hardware procurement, design programming, construction and commissioning, earning margins on engineering and product costs.
Typical clients: Enterprises with energy efficiency retrofit budgets.

5. Other Companies (Please research independently)

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